EXPERIMENT 8

Shravanya Andhale

D20A-02

Experiment 8: Semantic Search Using Embeddings and Vector Databases

AIM:

To perform semantic document retrieval by converting documents and user queries into vector embeddings and storing and searching those embeddings using a vector database such as FAISS or ChromaDB.

THEORY:

Semantic search is an information retrieval technique that finds relevant documents based on the meaning and context of a query rather than relying only on exact keyword matching. Traditional keyword-based search retrieves documents containing the same words as the user's query. However, it may fail when the user expresses an idea using different words or synonyms. Semantic search addresses this limitation by representing text as numerical vectors called embeddings.

1. Text Embeddings:

Text embeddings are numerical representations of text generated by an embedding model. Each document or text chunk is converted into a fixed-length vector containing information about its semantic meaning.

Texts with similar meanings tend to have vectors that are located close to each other in the learned vector space. For example, the sentences "How does machine learning work?" and "Explain the working of ML" have different words but similar meanings. An embedding model can represent these sentences with related vector representations.

2. Document Preprocessing and Chunking:

Before generating embeddings, documents are loaded and their text is extracted. Unnecessary formatting characters and extra spaces may be removed. Long documents are divided into smaller sections called chunks. Chunking makes retrieval more focused

because the system can return only the relevant portion of a document instead of retrieving the entire document.

Each chunk can be assigned a unique identifier and metadata such as document title, source, page number, section, and chunk ID. This information helps identify where the retrieved content originated.

3. Vector Similarity:

After document embeddings are generated, a user query is also converted into an embedding using the same embedding model. The query vector is compared with stored document vectors to identify the most similar results.

Cosine similarity is a commonly used measure for comparing vectors. It calculates the cosine of the angle between two vectors and indicates how closely they are aligned. A higher cosine similarity generally indicates greater semantic relatedness.

$$\text{Cosine Similarity} = \frac{A \cdot B}{\|A\|\|B\|}$$

Here, A and B represent the query and document embedding vectors.

4. Vector Databases:

Vector databases and vector indexes store numerical embeddings and support efficient similarity searches.

- FAISS: Facebook AI Similarity Search is a library used for efficient nearest-neighbor search over dense vectors. It allows embeddings to be indexed and searched quickly.
- ChromaDB: ChromaDB is a vector database that supports storing embeddings along with documents and associated metadata. It provides collection-based storage and retrieval.

5. Semantic Retrieval Process:

The process begins by loading documents, extracting their text, and splitting them into meaningful chunks. An embedding model converts each chunk into a vector. These vectors are stored in FAISS or ChromaDB. When a user submits a query, the same embedding model converts the query into a vector. The vector database compares the query vector with stored vectors and retrieves the top-k most similar chunks.

The retrieved results are displayed along with their document identifiers, metadata, similarity scores, or distance values, depending on the implementation.

The retrieval system can be evaluated using queries that closely match the original document wording, paraphrased queries, and queries whose answers are absent from the document collection. This helps measure whether the system retrieves meaningful information.

Semantic search forms an important part of Retrieval-Augmented Generation systems because it provides relevant document context to generative language models.

CODE AND OUTPUT:

Experiment 8 - Semantic Search using Embeddings and FAISS

```
!pip -q install sentence-transformers faiss-cpu pandas
```

```
# Small educational corpus based on the public BattingCoach README/project description.
# It summarizes project claims; it is not a substitute for the repository source or a coaching manual.
documents = [
 {"doc_id":"BC01","title":"Project overview","source":"BattingCoach README",
  "text":"BatCoach AI Pro is described as a browser-based cricket batting laboratory and virtual coach. It analyzes webcam video and provides feedback about batting technique."},
 {"doc_id":"BC02","title":"Pose and biomechanics","source":"BattingCoach README",
  "text":"The project describes MediaPipe-based pose estimation and kinematic angle extraction. Its biomechanics module evaluates cues such as lead elbow elevation, lead knee flexion, head-over-knee balance, and spine forward lean."},
 {"doc_id":"BC03","title":"Practice modes","source":"BattingCoach README",
  "text":"The README describes shadow practice without a bat and live willow practice with a bat. Shadow practice bypasses bat-specific checks. Live willow mode includes bat tracking."},
 {"doc_id":"BC04","title":"Computer vision and shot classification","source":"BattingCoach README",
  "text":"The project describes YOLOv8-OBB for oriented bat tracking in live willow mode and a VideoMAE transformer for classifying cricket shots across ten foundational shot types."},
 {"doc_id":"BC05","title":"Feedback and repeated errors","source":"BattingCoach README",
  "text":"The project describes on-screen coaching cues and voice feedback. It tracks total swings and clean repetitions, and repeated technique errors can trigger an intervention that freezes the rep counter."},
 {"doc_id":"BC06","title":"Limitations and responsible use","source":"Notebook-authored evaluation note",
  "text":"Computer-vision feedback depends on camera angle, lighting, occlusion, model confidence and the quality of the input video. A single recording cannot guarantee future match performance or diagnose a medical condition. Treat automated feedback as an aid for practi
]
chunks = []
for d in documents:
    # short chunks preserve the source title and unique chunk id
    chunks.append({"chunk_id":d["doc_id"]+"-C1","title":d["title"],"source":d["source"],"text":d["text"]})
print("Documents:",len(documents),"Chunks:",len(chunks))
for c in chunks: print(c["chunk_id"],"|",c["title"],"|",c["text"][:90]+"...")
```

```
Documents: 6 Chunks: 6
BC01-C1 | Project overview | BatCoach AI Pro is described as a browser-based cricket batting laboratory and virtual coa...
BC02-C1 | Pose and biomechanics | The project describes MediaPipe-based pose estimation and kinematic angle extraction. Its ...
BC03-C1 | Practice modes | The README describes shadow practice without a bat and live willow practice with a bat. Sh...
BC04-C1 | Computer vision and shot classification | The project describes YOLOv8-OBB for oriented bat tracking in live willow mode and a Video...
BC05-C1 | Feedback and repeated errors | The project describes on-screen coaching cues and voice feedback. It tracks total swings a...
BC06-C1 | Limitations and responsible use | Computer-vision feedback depends on camera angle, lighting, occlusion, model confidence an...
```

```
import numpy as np, pandas as pd, faiss
from sentence_transformers import SentenceTransformer
EMBED_MODEL = "sentence-transformers/all-MiniLM-L6-v2"
embedder = SentenceTransformer(EMBED_MODEL)
texts=[c["text"] for c in chunks]
embeddings=embedder.encode(texts,convert_to_numpy=True,normalize_embeddings=True).astype("float32")
print("Embedding model:",EMBED_MODEL)
print("Embedding shape:",embeddings.shape)
index=faiss.IndexFlatIP(embeddings.shape[1]) # inner product = cosine similarity after normalization
index.add(embeddings)
print("FAISS index size:",index.ntotal)
```

```
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
modules.json: 100% 349/349 [00:00<00:00, 25.4kB/s]
config_sentence_transformers.json: 100% 116/116 [00:00<00:00, 8.15kB/s]
README.md: 100% 10.5k/10.5k [00:00<00:00, 703kB/s]
sentence_bert_config.json: 100% 53.0/53.0 [00:00<00:00, 4.10kB/s]
config.json: 100% 612/612 [00:00<00:00, 48.5kB/s]
model.safetensors: reconstructing file: 100% 90.9MB / 90.9MB, 8.92MB/s
model.safetensors: downloading bytes: 85.0MB, 8.21MB/s
Loading weights: 100% 103/103 [00:00<00:00, 2505.69it/s]
tokenizer_config.json: 100% 350/350 [00:00<00:00, 26.4kB/s]
vocab.txt: 100% 232k/232k [00:00<00:00, 8.41MB/s]
tokenizer.json: 100% 466k/466k [00:00<00:00, 21.2MB/s]
special_tokens_map.json: 100% 112/112 [00:00<00:00, 7.39kB/s]
config.json: 100% 190/190 [00:00<00:00, 13.3kB/s]
Embedding model: sentence-transformers/all-MiniLM-L6-v2
Embedding shape: (6, 384)
FAISS index size: 6
```

```
queries = [
 "What technologies estimate body pose and joint angles?",
 "How does the system track the bat when a player uses one?",
 "What happens when the same technique mistake is repeated?",
 "Can the webcam guarantee a century in the next match?",
 "What is the difference between shadow practice and live willow mode?"
]
TOP_K=3
retrieval_rows=[]
for query in queries:
    qv=embedder.encode([query],convert_to_numpy=True,normalize_embeddings=True).astype("float32")
    scores, ids=index.search(qv,TOP_K)
    print("\nQUERY:",query)
    for rank,(score,idx) in enumerate(zip(scores[0],ids[0]),start=1):
        c=chunks[int(idx)]
        print(f"Rank {rank} | {c['chunk_id']} | cosine={score:.3f} | {c['title']}\n{c['text']}")
        retrieval_rows.append({"query":query,"rank":rank,"chunk_id":c["chunk_id"],"title":c["title"],
                               "cosine_similarity":float(score),"retrieved_text":c["text"]})
retrieval_df=pd.DataFrame(retrieval_rows)
display(retrieval_df)
```

```
QUERY: What technologies estimate body pose and joint angles?
Rank 1 | BC02-C1 | cosine=0.659 | Pose and biomechanics
The project describes MediaPipe-based pose estimation and kinematic angle extraction. Its biomechanics module evaluates cues such as lead elbow elevation, lead knee flexion, head-over-knee balance, and spine forward lean.
Rank 2 | BC06-C1 | cosine=0.322 | Limitations and responsible use
Computer-vision feedback depends on camera angle, lighting, occlusion, model confidence and the quality of the input video. A single recording cannot guarantee future match performance or diagnose a medical condition. Treat automated feedback as an aid for practice, not a
Rank 3 | BC04-C1 | cosine=0.283 | Computer vision and shot classification
The project describes YOLOv8-OBB for oriented bat tracking in live willow mode and a VideoMAE transformer for classifying cricket shots across ten foundational shot types.

QUERY: How does the system track the bat when a player uses one?
Rank 1 | BC01-C1 | cosine=0.535 | Project overview
BatCoach AI Pro is described as a browser-based cricket batting laboratory and virtual coach. It analyzes webcam video and provides feedback about batting technique.
Rank 2 | BC04-C1 | cosine=0.518 | Computer vision and shot classification
The project describes YOLOv8-OBB for oriented bat tracking in live willow mode and a VideoMAE transformer for classifying cricket shots across ten foundational shot types.
Rank 3 | BC03-C1 | cosine=0.476 | Practice modes
The README describes shadow practice without a bat and live willow practice with a bat. Shadow practice bypasses bat-specific checks. Live willow mode includes bat tracking.

QUERY: What happens when the same technique mistake is repeated?
Rank 1 | BC05-C1 | cosine=0.261 | Feedback and repeated errors
The project describes on-screen coaching cues and voice feedback. It tracks total swings and clean repetitions, and repeated technique errors can trigger an intervention that freezes the rep counter.
Rank 2 | BC06-C1 | cosine=0.250 | Limitations and responsible use
Computer-vision feedback depends on camera angle, lighting, occlusion, model confidence and the quality of the input video. A single recording cannot guarantee future match performance or diagnose a medical condition. Treat automated feedback as an aid for practice, not a
Rank 3 | BC03-C1 | cosine=0.120 | Practice modes
The README describes shadow practice without a bat and live willow practice with a bat. Shadow practice bypasses bat-specific checks. Live willow mode includes bat tracking.

QUERY: Can the webcam guarantee a century in the next match?
Rank 1 | BC06-C1 | cosine=0.378 | Limitations and responsible use
Computer-vision feedback depends on camera angle, lighting, occlusion, model confidence and the quality of the input video. A single recording cannot guarantee future match performance or diagnose a medical condition. Treat automated feedback as an aid for practice, not a
Rank 2 | BC01-C1 | cosine=0.298 | Project overview
BatCoach AI Pro is described as a browser-based cricket batting laboratory and virtual coach. It analyzes webcam video and provides feedback about batting technique.
```

```
QUERY: Can the webcam guarantee a century in the next match?
Rank 1 | BC06-C1 | cosine=0.378 | Limitations and responsible use
Computer-vision feedback depends on camera angle, lighting, occlusion, model confidence and the quality of the input video. A single recording cannot guarantee future match performance or diagnose a medical condition. Treat automated feedback as an aid for practice, not a
Rank 2 | BC01-C1 | cosine=0.298 | Project overview
BatCoach AI Pro is described as a browser-based cricket batting laboratory and virtual coach. It analyzes webcam video and provides feedback about batting technique.
Rank 3 | BC04-C1 | cosine=0.202 | Computer vision and shot classification
The project describes YOLOv8-OBB for oriented bat tracking in live willow mode and a VideoMAE transformer for classifying cricket shots across ten foundational shot types.

QUERY: What is the difference between shadow practice and live willow mode?
Rank 1 | BC03-C1 | cosine=0.775 | Practice modes
The README describes shadow practice without a bat and live willow practice with a bat. Shadow practice bypasses bat-specific checks. Live willow mode includes bat tracking.
Rank 2 | BC04-C1 | cosine=0.259 | Computer vision and shot classification
The project describes YOLOv8-OBB for oriented bat tracking in live willow mode and a VideoMAE transformer for classifying cricket shots across ten foundational shot types.
Rank 3 | BC01-C1 | cosine=0.158 | Project overview
BatCoach AI Pro is described as a browser-based cricket batting laboratory and virtual coach. It analyzes webcam video and provides feedback about batting technique.
```

	query	rank	chunk_id	title	cosine_similarity	retrieved_text
0	What technologies estimate body pose and joint...	1	BC02-C1	Pose and biomechanics	0.659236	The project describes MediaPipe-based pose est...
1	What technologies estimate body pose and joint...	2	BC06-C1	Limitations and responsible use	0.321618	Computer-vision feedback depends on camera ang...
2	What technologies estimate body pose and joint...	3	BC04-C1	Computer vision and shot classification	0.283450	The project describes YOLOv8-OBB for oriented ...
3	How does the system track the bat when a playe...	1	BC01-C1	Project overview	0.535487	BatCoach AI Pro is described as a browser-base...
4	How does the system track the bat when a playe...	2	BC04-C1	Computer vision and shot classification	0.517584	The project describes YOLOv8-OBB for oriented ...
5	How does the system track the bat when a playe...	3	BC03-C1	Practice modes	0.475838	The README describes shadow practice without a...
6	What happens when the same technique mistake i...	1	BC05-C1	Feedback and repeated errors	0.261472	The project describes on-screen coaching cues ...
7	What happens when the same technique mistake i...	2	BC06-C1	Limitations and responsible use	0.249859	Computer-vision feedback depends on camera ang...
8	What happens when the same technique mistake i...	3	BC03-C1	Practice modes	0.120093	The README describes shadow practice without a...
9	Can the webcam guarantee a century in the next...	1	BC06-C1	Limitations and responsible use	0.378362	Computer-vision feedback depends on camera ang...
10	Can the webcam guarantee a century in the next...	2	BC01-C1	Project overview	0.297917	BatCoach AI Pro is described as a browser-base...
11	Can the webcam guarantee a century in the next...	3	BC04-C1	Computer vision and shot classification	0.201571	The project describes YOLOv8-OBB for oriented ...
12	What is the difference between shadow practice...	1	BC03-C1	Practice modes	0.774776	The README describes shadow practice without a...
13	What is the difference between shadow practice...	2	BC04-C1	Computer vision and shot classification	0.258529	The project describes YOLOv8-OBB for oriented ...
14	What is the difference between shadow practice...	3	BC01-C1	Project overview	0.157873	BatCoach AI Pro is described as a browser-base...

```
judgements=retrieval_df.copy()
judgements["Relevance judgement"]=""
judgements["Reason"]=""
display(judgements)
judgements.to_csv("experiment8_retrieval_judgements.csv",index=False)
```

	query	rank	chunk_id	title	cosine_similarity	retrieved_text	Relevance judgement	Reason
0	What technologies estimate body pose and joint...	1	BC02-C1	Pose and biomechanics	0.659236	The project describes MediaPipe-based pose est...		
1	What technologies estimate body pose and joint...	2	BC06-C1	Limitations and responsible use	0.321618	Computer-vision feedback depends on camera ang...		
2	What technologies estimate body pose and joint...	3	BC04-C1	Computer vision and shot classification	0.283450	The project describes YOLOv8-OBB for oriented ...		
3	How does the system track the bat when a playe...	1	BC01-C1	Project overview	0.535487	BatCoach AI Pro is described as a browser-base...		
4	How does the system track the bat when a playe...	2	BC04-C1	Computer vision and shot classification	0.517584	The project describes YOLOv8-OBB for oriented ...		
5	How does the system track the bat when a playe...	3	BC03-C1	Practice modes	0.475838	The README describes shadow practice without a...		
6	What happens when the same technique mistake i...	1	BC05-C1	Feedback and repeated errors	0.261472	The project describes on-screen coaching cues ...		
7	What happens when the same technique mistake i...	2	BC06-C1	Limitations and responsible use	0.249859	Computer-vision feedback depends on camera ang...		
8	What happens when the same technique mistake i...	3	BC03-C1	Practice modes	0.120093	The README describes shadow practice without a...		
9	Can the webcam guarantee a century in the next...	1	BC06-C1	Limitations and responsible use	0.378362	Computer-vision feedback depends on camera ang...		
10	Can the webcam guarantee a century in the next...	2	BC01-C1	Project overview	0.297917	BatCoach AI Pro is described as a browser-base...		
11	Can the webcam guarantee a century in the next...	3	BC04-C1	Computer vision and shot classification	0.201571	The project describes YOLOv8-OBB for oriented ...		
12	What is the difference between shadow practice...	1	BC03-C1	Practice modes	0.774776	The README describes shadow practice without a...		
13	What is the difference between shadow practice...	2	BC04-C1	Computer vision and shot classification	0.258529	The project describes YOLOv8-OBB for oriented ...		
14	What is the difference between shadow practice...	3	BC01-C1	Project overview	0.157873	BatCoach AI Pro is described as a browser-base...		

https://colab.research.google.com/drive/1DN4yEYUrdoj7SRQgWtY0-DVrkyTWmsId?usp=sharing

CONCLUSION:

The experiment demonstrated semantic document retrieval using text embeddings and vector search. Documents were converted into numerical vectors and stored using a vector database. User queries were embedded using the same model, and relevant document chunks were retrieved based on vector similarity. The experiment showed how semantic search can identify related information even when the query uses different wording from the original documents.